Audio Recorder - Part 1

04

Sound waves can be digitally replicated to generate any sound available to us. The first step to digitally enhancing or creating a sound is to get access to the original audio/sound. An audio recording technique provides a great way for us to listen to a voice repeatedly, in the form of an audio recording. Applications used to record an audio are commonly available on the platforms that are offline, or web-based, i.e., online.

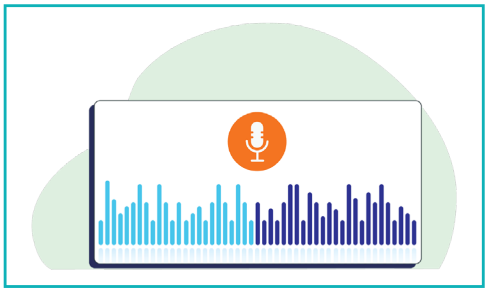
‘**JavaScript**’ is the most used programming language for developing web-based applications. Social media platforms, like Facebook, Instagram, other dynamic websites, and web-based games, all use JavaScript extensively. In this lesson, we will explore how an audio recording app can be created using JavaScript in p5js.

An **'audio recorder'** can capture sound using a microphone and save the recorded sound in the form of a digital audio file.

There are separate electronic devices to record the audio. But we can also use an audio recorder app on smart phones. It is a simple utility app that serves the exact same purpose. Different audio recording apps can be found on all platforms, such as the web, mobile app stores, etc. They can be downloaded on your PC, mobile phone, and tablet.

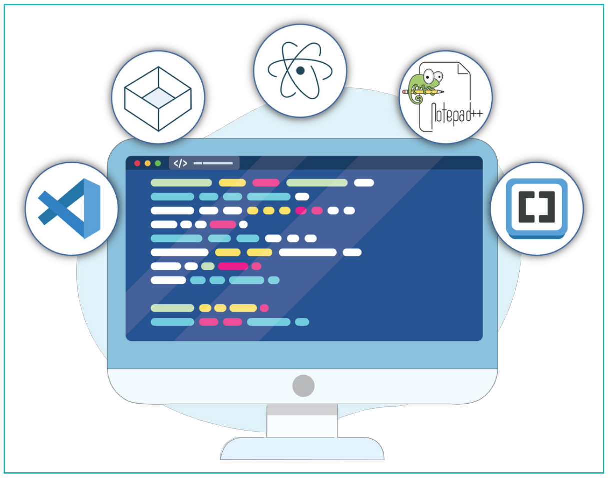
Example:
'Voice Recorder' app available on Google Play.

In this lesson, we will create a web-based application to record audio using the 'p5js library'. So far, we have used the p5js online editor. But we can also create applications offline using a code editor. This is how professional coders mostly work.



Code Editor

The **'Code editor'** is an optimized text editor specifically designed to code in different programming languages with the necessary features for coding. Some of these features include compilers for development in various programming languages, code debugging, syntax highlighting, code assist, auto-completion, compilation and code execution, output display, etc.



The p5js code editor enables us to access the code area, output preview, and console within a single browser window. However, it can only be used to save a project using an active internet connection.

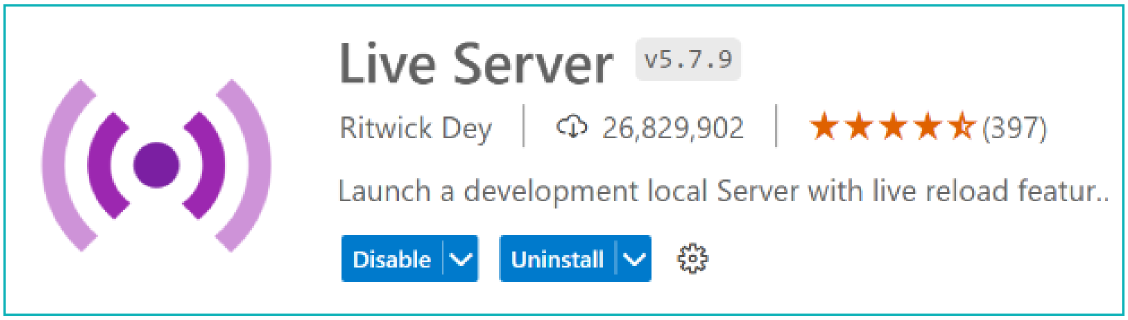
We also can create a stand-alone web-based application independently without a p5js editor interface. Therefore, an offline code editor software is required which can be used to write, edit, and compile the code and check the output.

Any open-source code editor can be used here. Some examples are VS Code, Atom, Sublime Text, etc.

In this session, we will use the 'VS Code' editor:

- Search for 'Visual Studio Code Download' on Google and click on the download link or type https://code.visualstudio.com/download
- Install the 'VS Code editor' suitable for your operating system from the download page.

Note:
When the VS Code is installed, install an extension called 'live server' from the 'extensions' option. It is used to see the output in the browser.

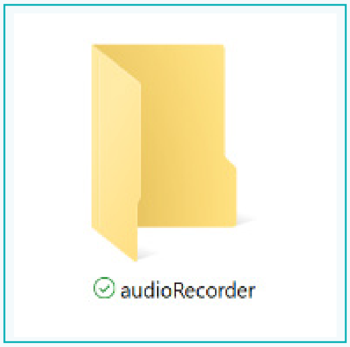


Activity

Step 1: Project Setup

All the files created in the p5js file structure can be saved in a single project folder.

1. Create a project folder named “audioRecorder” at a suitable location on your computer. All files related to the project will be saved in this folder.



The p5js online code editor file structure contains multiple files such as ‘index.html’, ‘style.css’, and ‘sketch.js’. A default code in index.html in the p5js web editor is used to link p5js libraries and other files together to show the output.

index.html

```html
<!DOCTYPE html>
<html lang="en">
  <head>
    <script src="https://cdnjs.cloudflare.com/ajax/libs/p5.js/1.3.1/p5.js"></script>
    <script src="https://cdnjs.cloudflare.com/ajax/libs/p5.js/1.3.1/addons/p5.sound.min.js"></script>
    <link rel="stylesheet" type="text/css" href="style.css">
    <meta charset="utf-8" />

  </head>
  <body>
    <script src="sketch.js"></script>
  </body>
</html>
```

sketch.js

```javascript
function setup() {
  createCanvas(400, 400);
}

function draw() {
  background(220);
}
```

style.css

```css
html, body {
  margin: 0;
  padding: 0;
}
canvas {
  display: block;
}
```

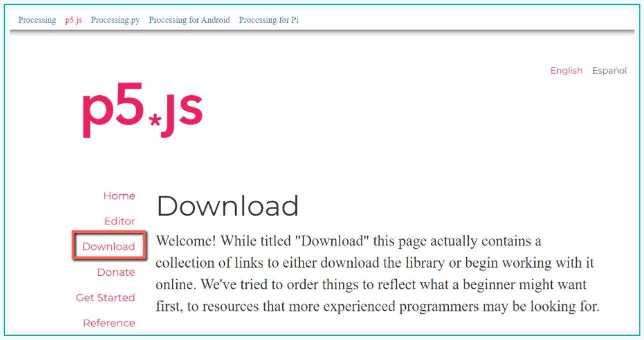
We want to use the p5js library commands and functions in our project even though we are using the external code editor.

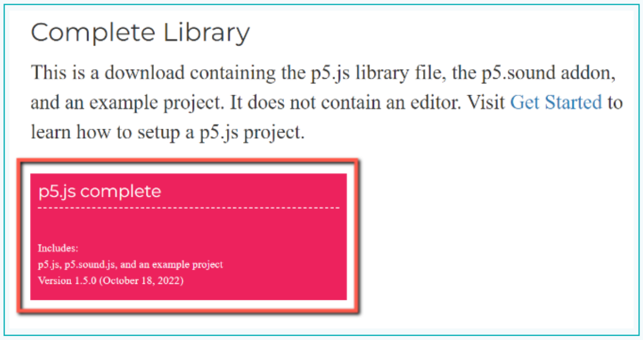
In case of coding in an offline editor, we need to download the p5js library files, include all these files in the project folder, and link them to the ‘index.html’ file.

2. Click on the ‘Downloads’ option on the ‘p5js.org’ page.

Activity

3. Go to the ‘Complete Library’ title and click on the box ‘p5.js complete’ to download the library files in a zipped format named ‘p5’.





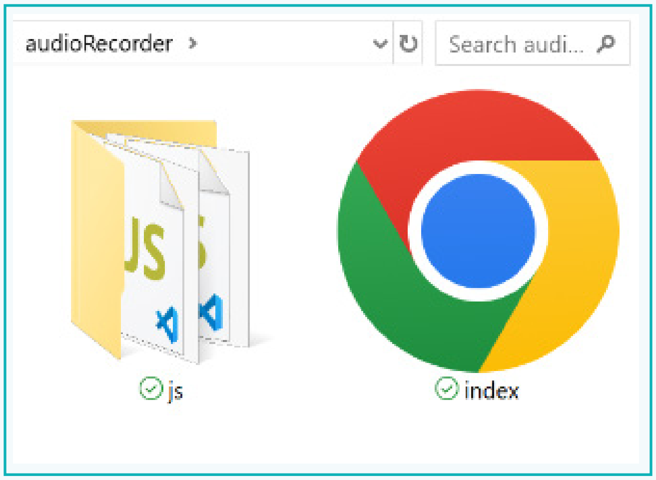
4. Extract the downloaded file to get a folder named ‘p5’.

5. Copy and paste the ‘index.html’ file from the folder ‘empty-example’ into the project folder ‘audioRecorder’.

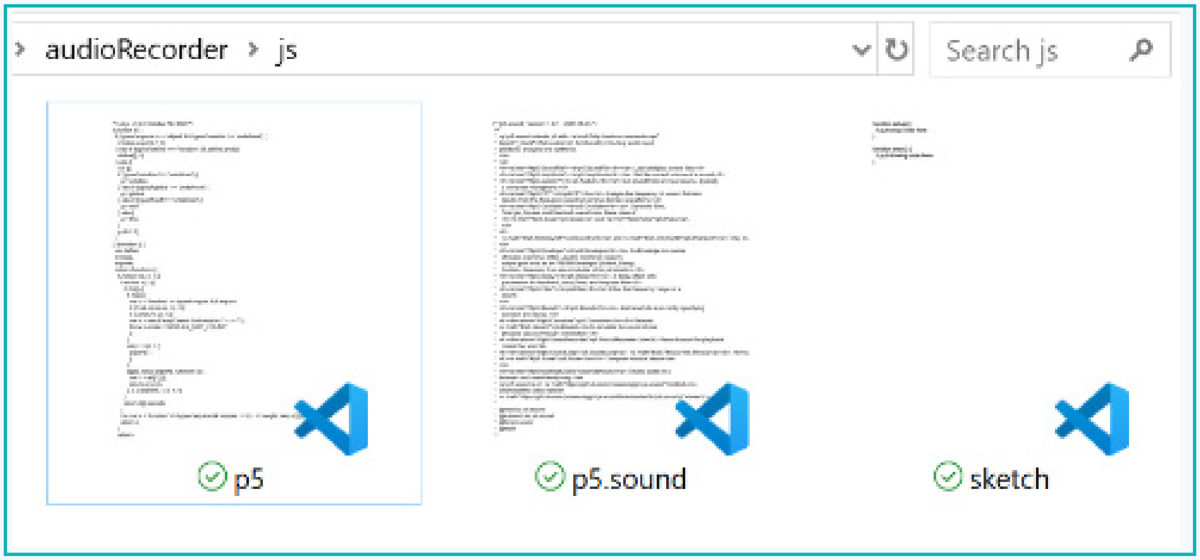
6. Create a folder named ‘js’ within the ‘project folder ‘audioRecorder’.

7. Copy the ‘p5.js’ file from the ‘lib’ folder and the ‘p5.sound.js’ file from the ‘addons’ folder, and paste into the newly created JS folder.

7. Copy and paste the sketch.js file from the folder ‘empty-example’ into the ‘js’ folder. This way, all JavaScript files are stored within the folder named ‘js’.



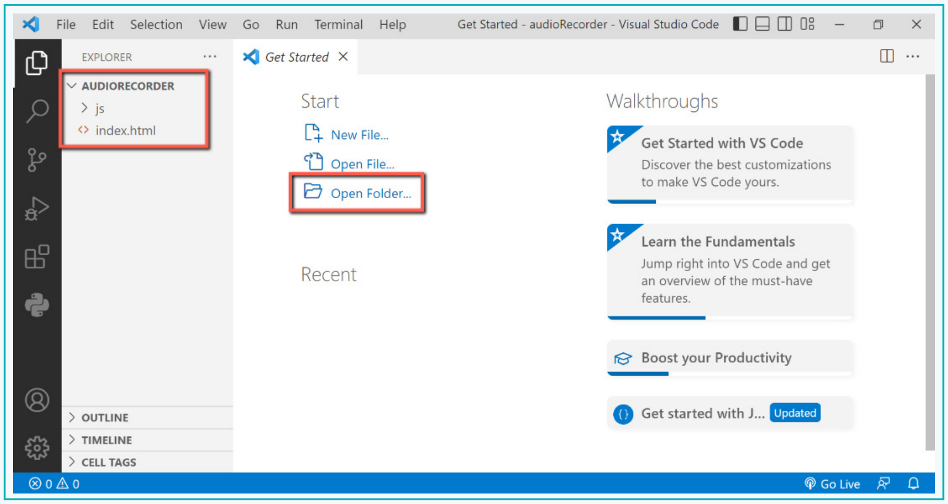
Activity



Now, the project folder is ready. We need to edit the code for linking the project files based on the file structure we created.

Step 2: Edit the Code

1. Click on the 'VS Code' icon to open the software. We will see the code editor interface.
2. Click on the 'Open Folder' option and add the folder we created for the project in the code editor.



Activity

We can now see the folder and the files in the Explorer window on the left.

3. Click on the ‘index.html’ file in the ‘EXPLORER’ window.

4. Observe the default code within the style tag for CSS, which we can move to a separate CSS file.

5. Observe the default code in the ‘index.html’ file that links to the JavaScript files such as ‘p5.js’, ‘p5.sound.js’, and ‘sketch.js’. We need to update the path according to the file structure.

6. Click on the ‘new file’ option in the explorer and create a new file named “style.css”.

7. Copy the code within the style tag in the ‘index.html’ file and then, paste it into the style.css file. Delete the code line for the background colour since we want the background to be blank.

8. Remove the ‘style’ tag from the ‘index.html’ file.

9. In index.html, uncomment the code by removing <!-- and --> from the code on the highlighted section for linking the ‘p5.sound.js’ file.

Activity

10 Update the path of linking all JavaScript files as “js/filename”.

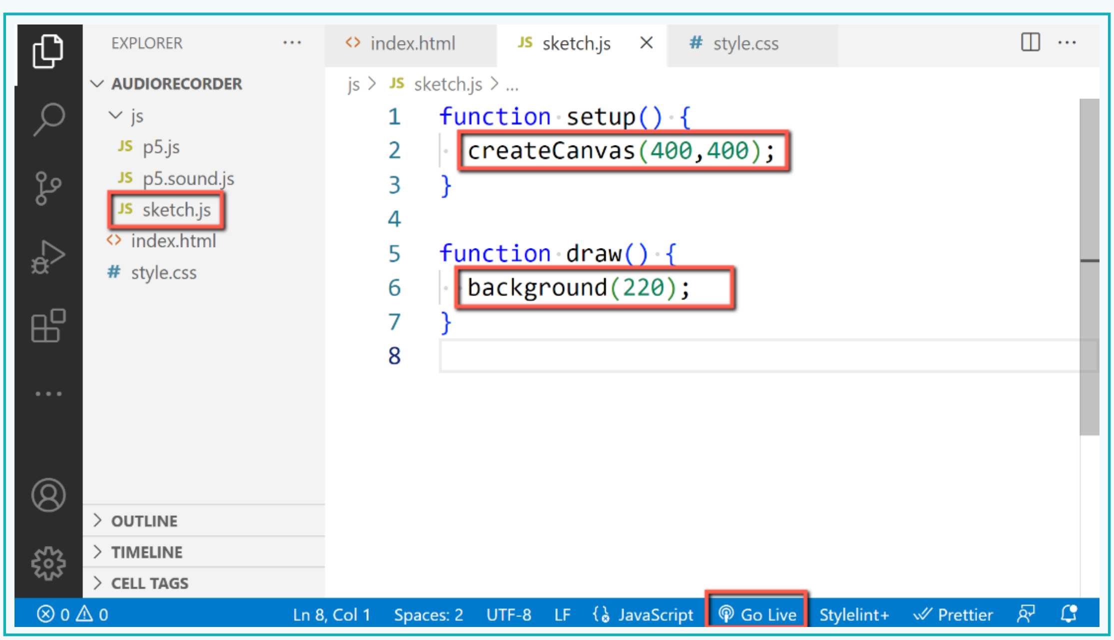
11 Change the name of the title tag to “Audio Recorder”.

12 Open the ‘sketch.js’ file.

13 Add basic code to create a ‘400x400 pixels’ canvas in the setup.

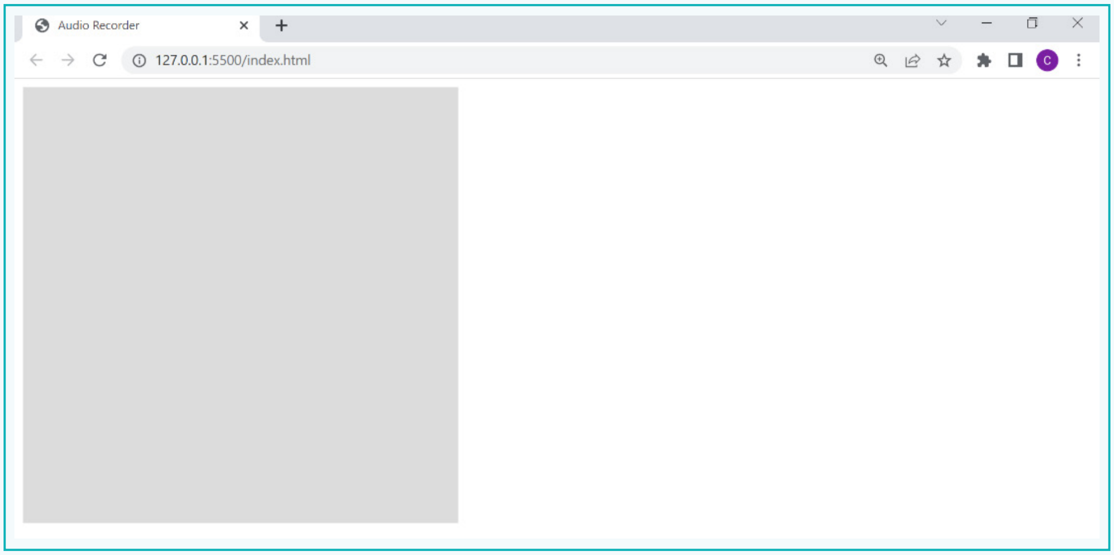
14 Add code to set the background colour of the canvas of greyscale value ‘220’ in draw.

15 Click on the ‘go-live’ option at the bottom toolbar to check the output in the default browser window.



Activity

The ‘Live Server’ extension should be installed to see the option ‘Go Live’ and to check the output. The output can be seen as follows.



We now have a coding area and output preview for our project. But we will also need a console that we have used so far for debugging the code. Let’s add it.

Exercise

Fill in the blanks.

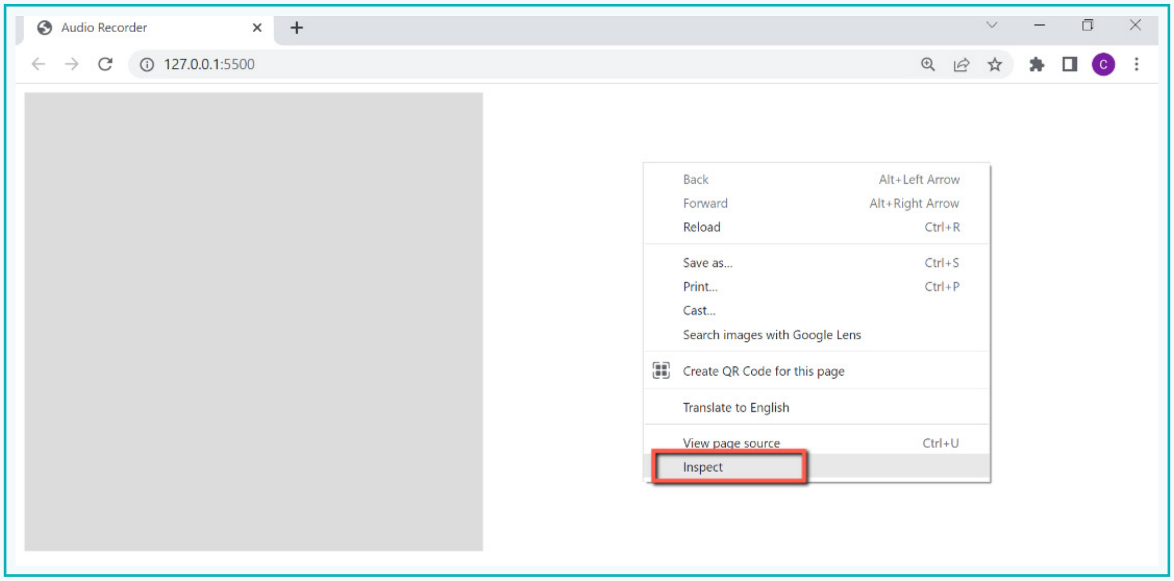
1. The JavaScript library files that we need to link to the index.html for using the p5js library commands and functions are ________________ and ________________.
2. ________________ is a text editing software with special features related to coding.

'Debugging' is the process of identifying and removing errors from the code. It is a very important part of coding. We use a console in the p5js editor to debug. We can also open the console in the browser while using the code editor.

Activity

Step 3: Debug the Code

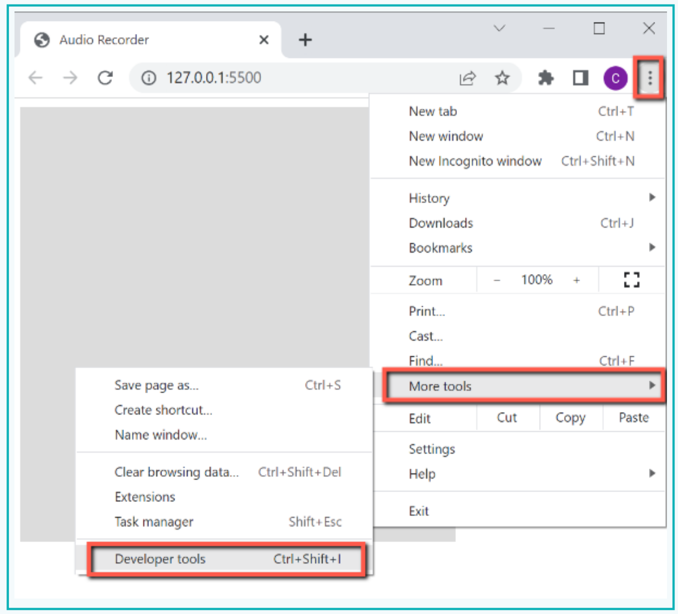
1. Right-click on the project output screen in the browser.
2. Click on the 'Inspect' option.



OR

1. Open the developer tools by clicking on the 3 vertical dots given in the right-hand corner. This will open browser options.

Activity

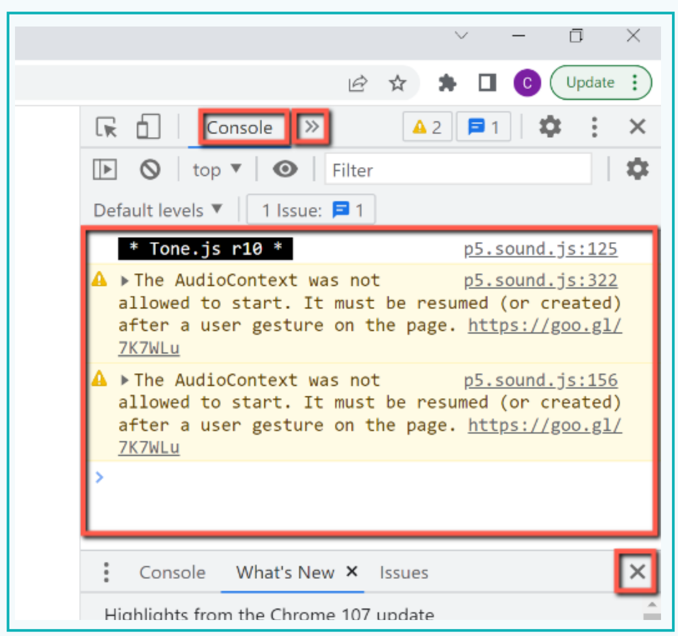


2. Select 'More tools' and click on the 'Developer tools' option.

A new section of the Developer tools will appear in the browser. This section contains many tools that can be used by developers for debugging a web-based project. The Console is one of those tools.

3. Click on the double-arrowheads at the top.

4. Select the 'Console' option.

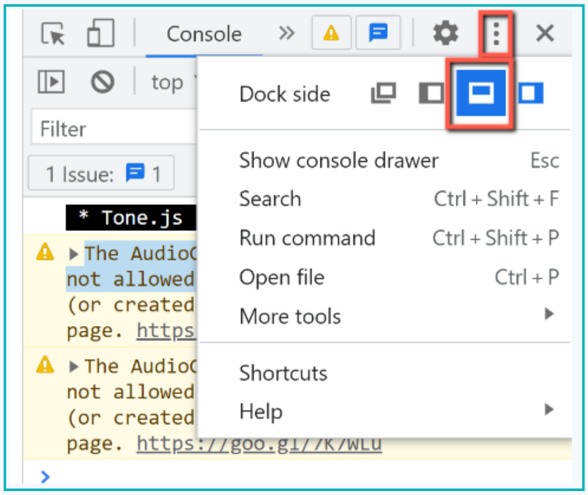


5. Close the other window, given on the lower half of this section by clicking on the 'x'.

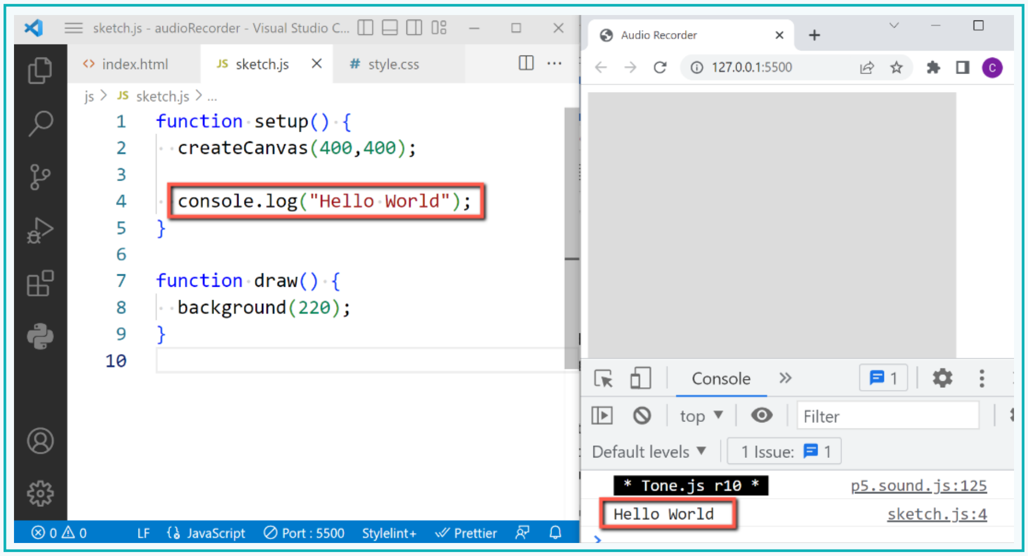
Activity

Now, we can see the console window in the browser. There may be some warnings due to some JavaSript library files like p5.sound.js on the console. These warnings are based on some browser policies related to audio libraries and can be ignored.

We can switch between the code editor window and the browser to check the intermittent output. We can also configure the output window with the console alongside the code editor. In this case, the console section can be shifted to the bottom side of the browser window by clicking on the three dots at the top left and select the option as shown.



6. Resize the code editor window, so they can be stacked side by side.
7. Add code console.log(“Hello World”); in the setup to check if it is working.



The console is now configured for debugging.

We have now completed the basic setup for an offline p5js project using ‘Visual Studio Code’. This can be copied and pasted as a reference folder. We can start with this copy of the reference folder any time we want to create a new project.

Now, the basic project setup has been completed. We can now begin coding to create an audio recorder.

4. Audio Recorder - Part 1

Exercise

Answer the question with one sentence

3. How will you open a console in a browser? Find out the procedure for the browser you are currently using.

▲ ____

____

Activity

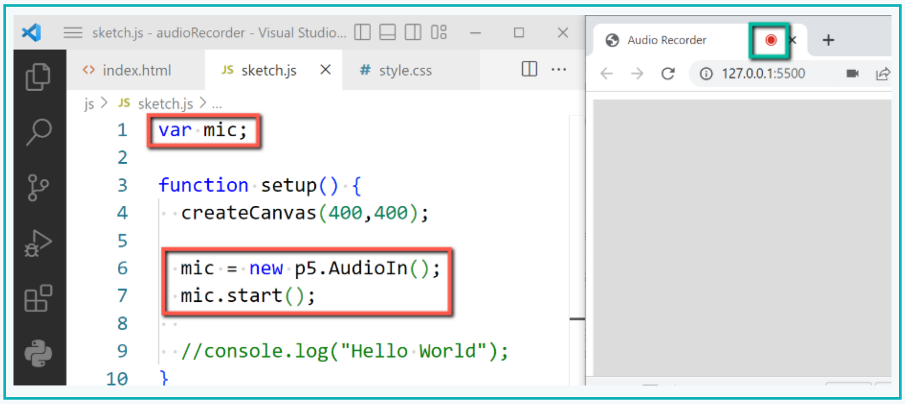
Step 4: Create an Audio Recorder

A mic is required to capture and receive an audio signal on an electronic or digital device.

We need to add a code to access the mic of the device, such as a laptop or a computer. Once we access the input from the mic, we can record the audio and add the data to a blank file. Then the file can be saved. These steps will be executed using the p5 sound library.

1. Create a new variable 'mic' at the beginning of the code.
2. In setup, add and create a new object of the 'p5.AudioIn()' class for getting input from microphone and assign it to mic variable.
3. Use the 'mic.start()' method to start the device's microphone.

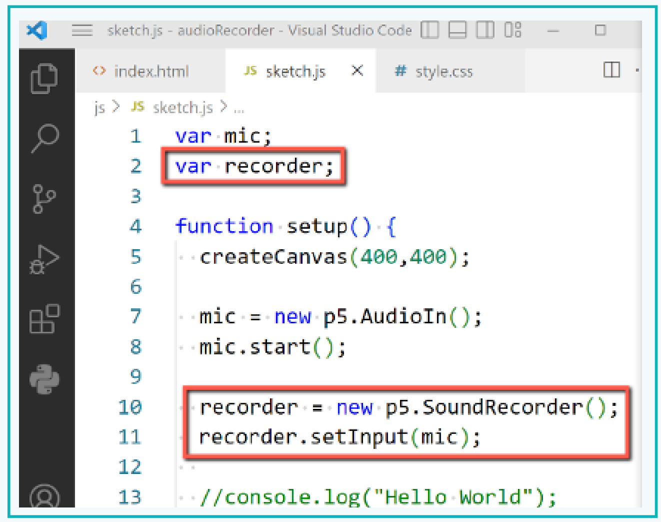
We will be able to see a red dot on the browser tab showing mic is on. Note that browser may ask us for the permission to use mic for the first time. We have to select 'allow' option to be able to use mic in our app.



Activity

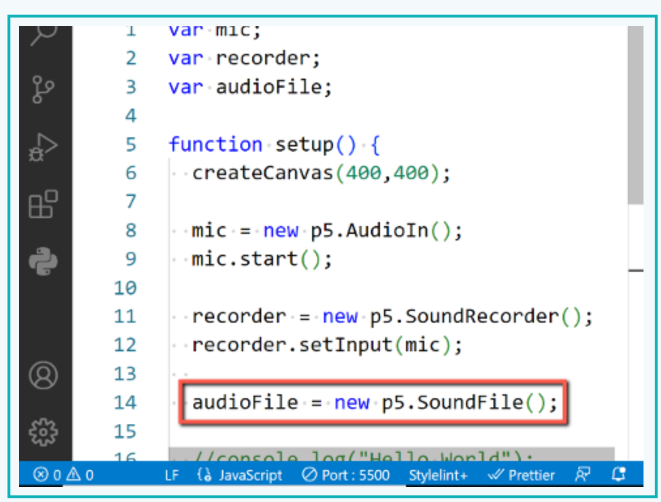
To be able to record the input from the mic, we need to create a new instance of a soundRecorder class from the p5 sound library.

4. Create a variable named “recorder” at the beginning of the code.
5. In the setup, create an object of the p5.soundRecorder class and assign it to the ‘recorder’ variable.
6. Set the ‘mic’ object created as an input to the ‘recorder’ using the ‘setInput()’ method.



Once the recording is done, we must save it as an audio file. We can create an empty file by creating a new instance of a ‘SoundFile’ class from the ‘p5 sound library’.

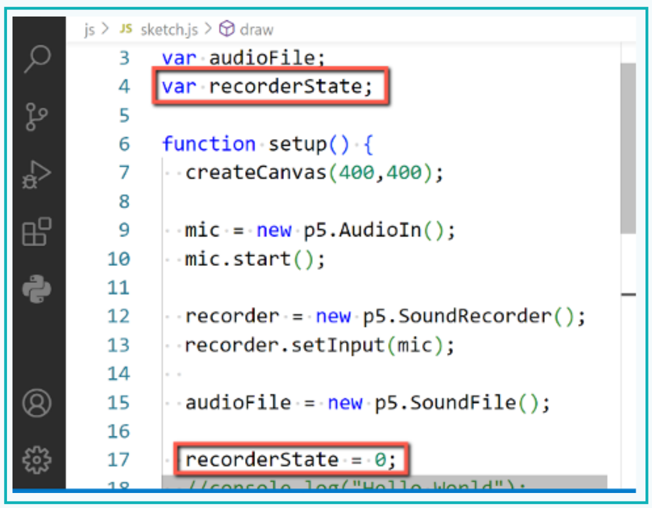
7. Create a variable named “audioFile” at the beginning of the code.
8. In the setup, create an instance of the ‘SoundFile()’ class of the p5 library and assign it to the ‘audioFile’ variable.



Activity

Now, with this setup, we can add functionality to the recording. We will create a status variable that can keep track of the recorder state.

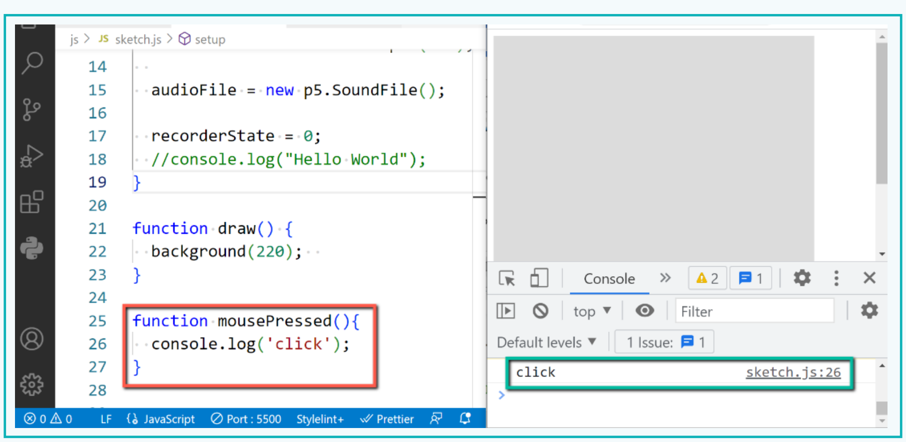
9. Create a variable named “recorderState” and initialize its value to ‘0’ in the setup to represent the recorder being ‘off’.



Based on the recorder state, we can perform actions such as start recording, stop recording, or save the file with a mouse click.

10. Add the ‘mousePressed()’ function to detect a mouse click.

11. Use the ‘console.log()’ command to check if the function is working.

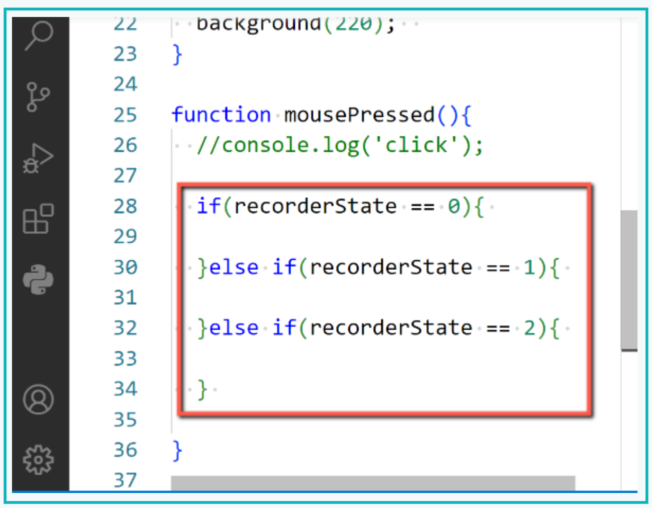


We can perform recorder actions based on the mouse click event by checking the value in the recorderState variable using the if else conditions.

Activity

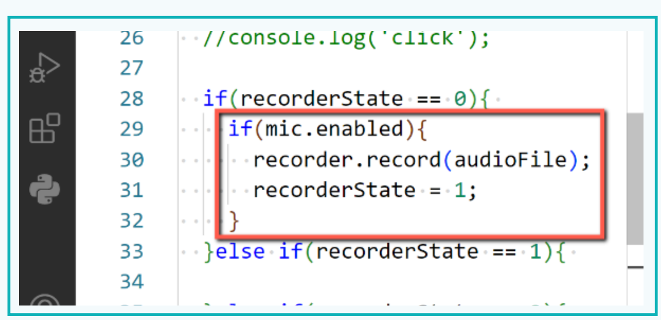
Step 5: Apply the ‘if else’ condition

1. Add the ‘if’ condition statement to check if the value of the ‘recorderState’ variable is ‘0’, indicating that the recorder is off. If ‘true’, we can start recording.
2. Add a second condition, using ‘else if’ to check if the value of the ‘recorderState’ variable is ‘1’, i.e., the recording is on. If ‘true’, we can stop the recording.
3. Add a third condition, using ‘else if’ to check if the value of the ‘recorderState’ variable is ‘2’, i.e., the recorded file can be saved. If ‘true’, we can save the recording in the sound file.



To start recording, we need to check if the recorder is stopped, i.e., if ‘recorderState’ is ‘0’. Also, check if the mic is turned on before recording.

4. Within the block of the condition if the ‘recorderState’ is ‘0’, add the ‘if’ condition to check if the mic is enabled before we started recording.
5. Add the ‘recorder.record(audioFile)’ command to start recording.
6. Change the ‘recordState’ variable value to ‘1’ to state that recording is on.

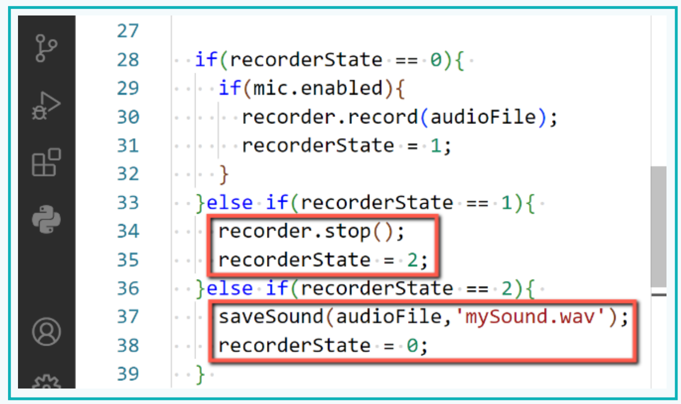


Activity

7. Add the code 'recorder.stop()' to the code block if the 'recorderState' value is '1', the recorder is 'on'. Therefore, if mouse button is clicked again when recording is on, it will be stopped.

8. Change the 'recorderState' variable value to '2' to state that recording is off.

9. In the condition block that checks if the 'recorderState' is '2', add the 'saveSound()' command. The input should be 'audioFile', and the name of the file to be saved, separated by a comma.



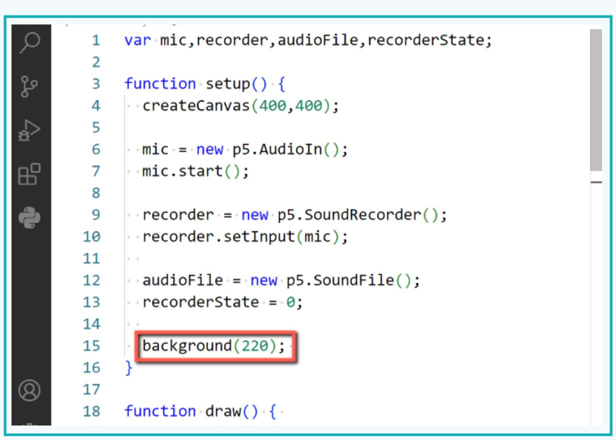
Now, we can start recording audio, stop recording, and save the file with the click operation. To identify different states, we can change the background colour.

Step 6: Change the Background Colour

1. Move the 'background(220)' command from the 'draw()' to 'setup()' to set the initial background colour only once at the beginning.

2. Change the background colour to red when the recording starts and to green when the recording ends.

3. Change the background colour back to the starting colour of greyscale '220', when the user clicks on 'Save the file'.



Activity

Run the code to check the working of the audio recorder. Sometimes, an error may appear on the console when a recording is stopped. We will not be able to save our recordings due to this error.

Web browsers, such as those on iOS and Google, enforce a policy, which mandates that no audio-related operations can occur until a user interaction event, like a mouse press, is initiated. As a result, the 'AudioContext' remains suspended until the user interacts with the webpage.

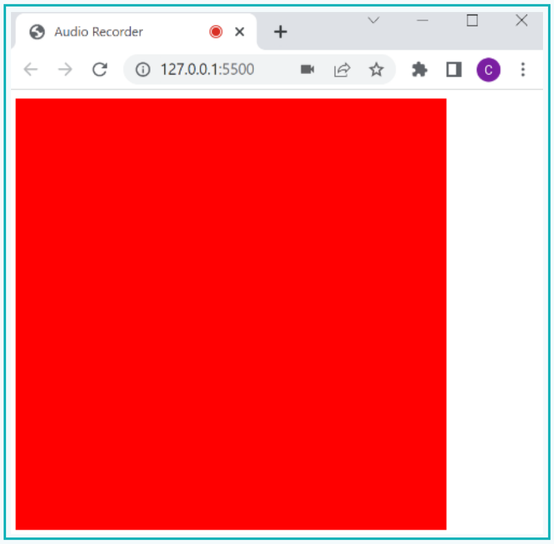
Note:
To use, play, modify, and control audio on web platforms, the 'AudioContext' object is used as designed by the web audio API. An audio context is an object that allows developers to use any audio-related features on the web.

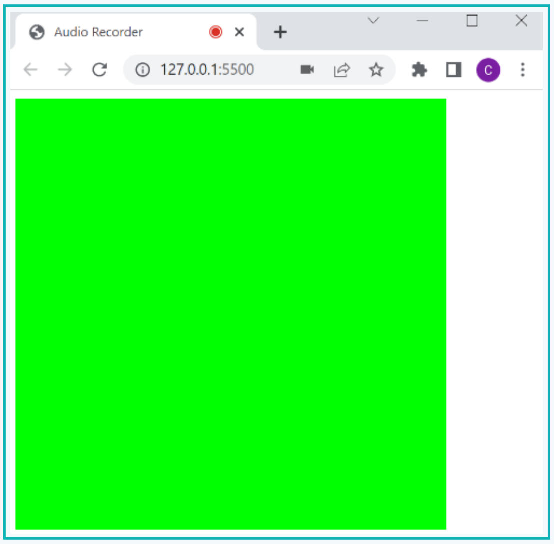
4. To save the recorded audio without error, we need to start an 'AudioContext' calling the 'userStartAudio()' command in mousePressed().

```
function mousePressed(){
  userStartAudio();
  //console.log('click');
  if(recorderState == 0){
    if(mic.enabled){
      recorder.record(audioFile);
```

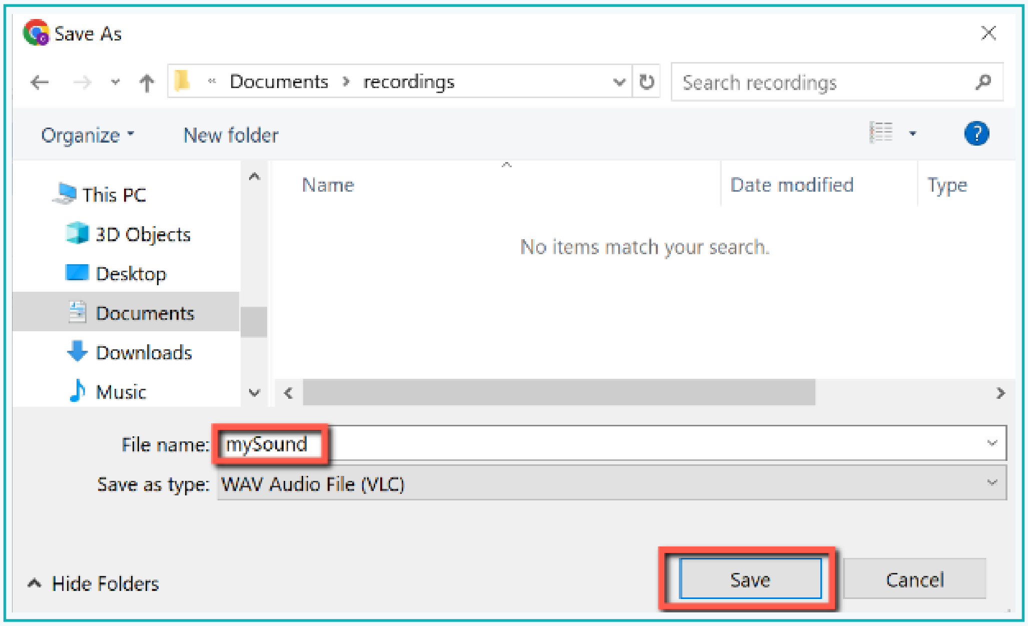
5. Run the code and check the working of the recorder.

6. Note that the red dot visible in the tab header, starts blinking when the recording starts.

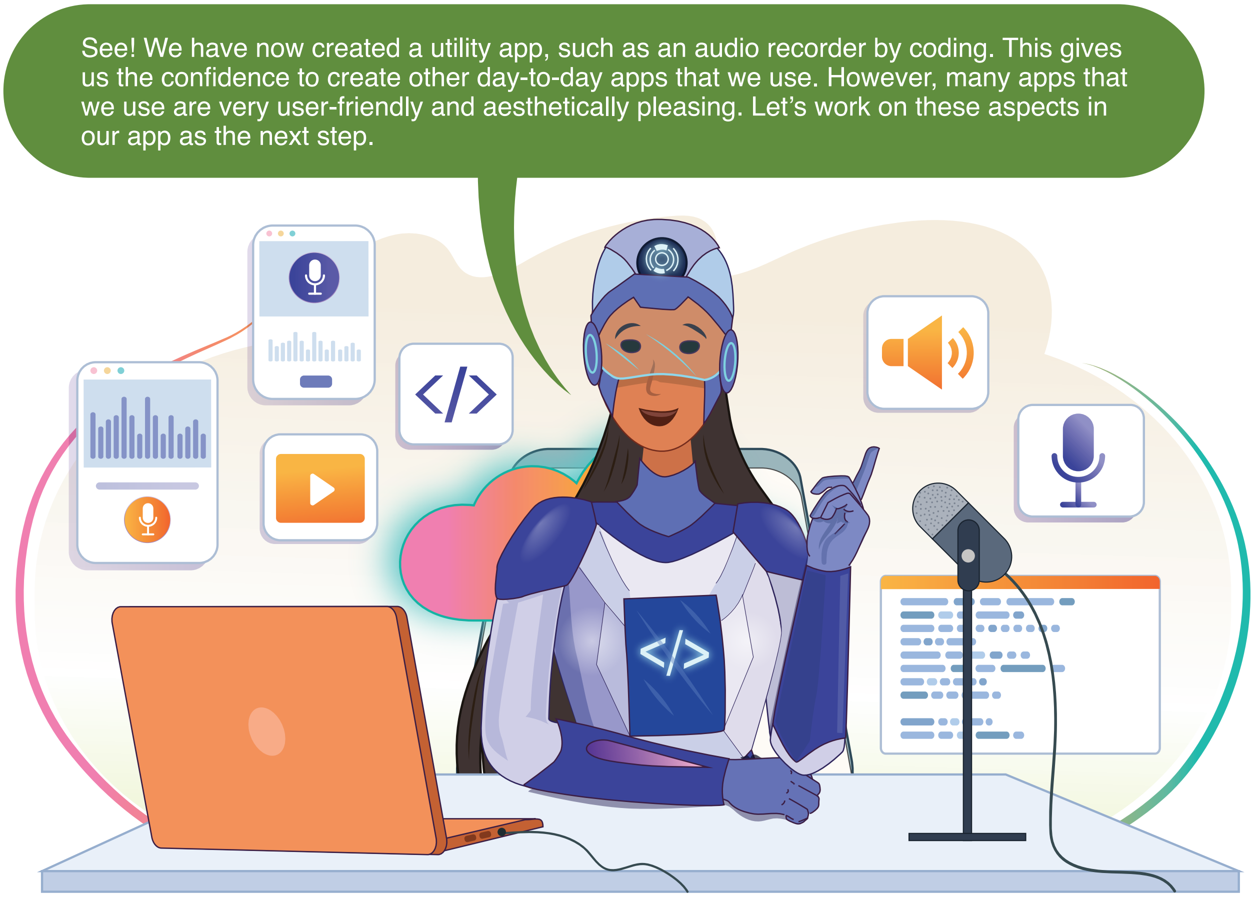




Activity



We are now familiar with the basic functionality of an audio recorder and how to create one. However, this application is not user-friendly and aesthetically pleasing. In the next lesson, we will be focusing on creating a user-centric audio recording app.



Exercise

State if the statements are True or False

4. The 'record()' method accepts an instance of 'p5.SoundFile' as an input parameter.

▲ ______________________

5. To stop recording, the 'stop()' method is used.

▲ ______________________

6. The 'SaveSound()' command is used to save the recorded sound as an audio file and requires only one parameter.

▲ ______________________

4. Audio Recorder - Part 1

Tech Flash

- **Code Editor** is an optimized text editor specifically designed for coding with the necessary features.
- **Developer Tools** refers to web development tools that are integrated into browsers and can be used to test and debug their code.
- **Mic** or **Microphone** is a device that converts audio signals into electrical signals, which are used to capture audio input.
- **Audio Recorder** is a tool or device that is used to record audio captured by a microphone and save it as an audio file.
- **WAV** or **Waveform Audio File Format** is an audio file format used for storing digital audio.